Lesson notes · for you

Sequences and the memory that fades

University, 3

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- The state is the memory
- Empty steps still multiply it

Activity: In the lab

1. Slider to the left
2. Slider to the right

Discussion

- What would a longer gap do?

Check yourself

- Does the end still know the cue? Write the number.

Try it this week
Write the hidden state after a cue of 1 and one empty step, if memory is one half.

Worksheet

Sequences and the memory that fades

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Name _____ Class _____ Date _____

1 **Memory 0.5. The cue enters the state as 1, then two empty steps follow. What is left?**

◯ 0.5 ◯ 0.25 ◯ 0.125

2 **A recurrent net reads the whole sequence in one look.**

◯ True ◯ False

3 **Match the word to the meaning.**

Hidden state	• • • •	The symbol that must still matter at the end
Cue	• • • •	A small memory times the state, again and again
Fading	• • • •	What the net carries forward

4 **Order one recurrent step.**

☐ Mix it with the old hidden state

☐ Keep the mix as the new state

☐ Read the next symbol

✎ **What did the empty steps do to the cue?**

Check your answers

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1 **0.25**

Each empty step multiplies the state by 0.5: 1 × 0.5 × 0.5 = 0.25.

2 **False**

It reads one symbol at a time and hopes the hidden state kept the rest.

3 **Hidden state — What the net carries forward**
Cue — The symbol that must still matter at the end
Fading — A small memory times the state, again and again

The lab's end number is the cue after it has been multiplied by the memory at every empty step.

4 **Top to bottom: 2, 3, 1**

The new state is the only memory the next symbol will see.